

Algebraic and Transcendental functions

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Algebra skills

Use the complete the squared method to find the standard form of the the following conic sections,

$$\begin{aligned}(a) \quad & 4x^2 + 9y^2 - 8x + 18y = -12 \\(b) \quad & 5x^2 + 5y^2 - 60x - 20y + 40 = 0 \\(c) \quad & -16x^2 + 9y - 128x + 18y - 391 = 0\end{aligned}$$

Function analysis

Find the inverse function of the following function and find their domain and range,

$$(a) \quad f(x) = \sqrt{5-2x}, \quad (b) \quad g(x) = \frac{1}{x-2} \quad (c) \quad h(x) = \left(\frac{1}{2}\right)^{-x} + 2$$

Modelling with math

Compound interest. A total of \$3000 is invested at an annual interest of 5.4%.

1. Write the model of this situation.
2. Find the balance after 3 years if it is compounded:
 - Annually
 - Monthly
 - Daily
3. If the interest is compounded semiannually, how long will it triple your money?

pH. In the chemistry world, pH is used to describe the acidity and basicity of a substance through the equation $pH = -\log H$, H being the concentration of hydrogen ions in moles per liter. Jeffrey knows that avocados have pH of 6.5, and chocolate has a pH of 5.4. Based on this information, which of these two has a lower concentration of hydrogen ions?

Algebra Toolkit

Algebra properties

$$\begin{aligned}a + b &= b + a, \\(a + b) + c &= a + (b + c), \\a(b + c) &= ab + ac, \\\frac{a}{b} + \frac{c}{d} &= \frac{ad + bc}{bd}, \quad \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}, \\\frac{-a}{b} &= -\frac{a}{b} = \frac{a}{-b}, \\\frac{\frac{a}{b}}{\frac{c}{d}} &= \frac{ad}{bc} \\\sqrt{ab} &= \sqrt{a}\sqrt{b}, \quad \sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}, \quad \sqrt[n]{a} = a^{1/n} \\x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}, \\a^2 - b^2 &= (a - b)(a + b)\end{aligned}$$

Laws of exponents

$$\begin{aligned}a^r \cdot a^s &= a^{r+s}, \quad (a^r)^s = a^{rs} \\\frac{a^r}{a^s} &= a^{r-s}, \quad (ab)^r = a^r b^r \\\left(\frac{a}{b}\right)^r &= \frac{a^r}{b^r}, \quad b \neq 0\end{aligned}$$

Laws of Logarithms

$$\begin{aligned}\log_a(AB) &= \log_a(A) + \log_a(B) \\\log_a\left(\frac{A}{B}\right) &= \log_a(A) - \log_a(B) \\\log_a(A^C) &= C \log_a(A)\end{aligned}$$

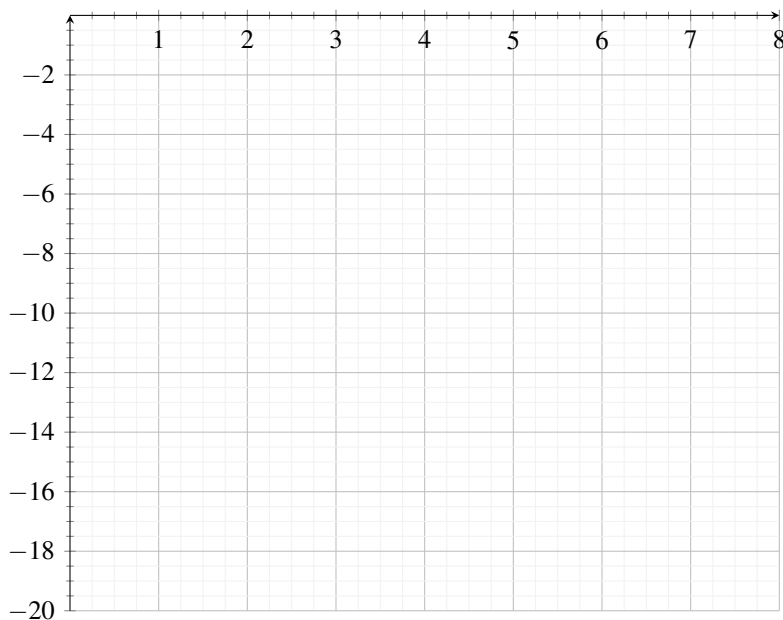
Inverse functions property

$$\begin{aligned}(f \circ f^{-1})(x) &= x = (f^{-1} \circ f)(x) \\\log_a(a^x) &= x = a^{\log_a(x)}\end{aligned}$$

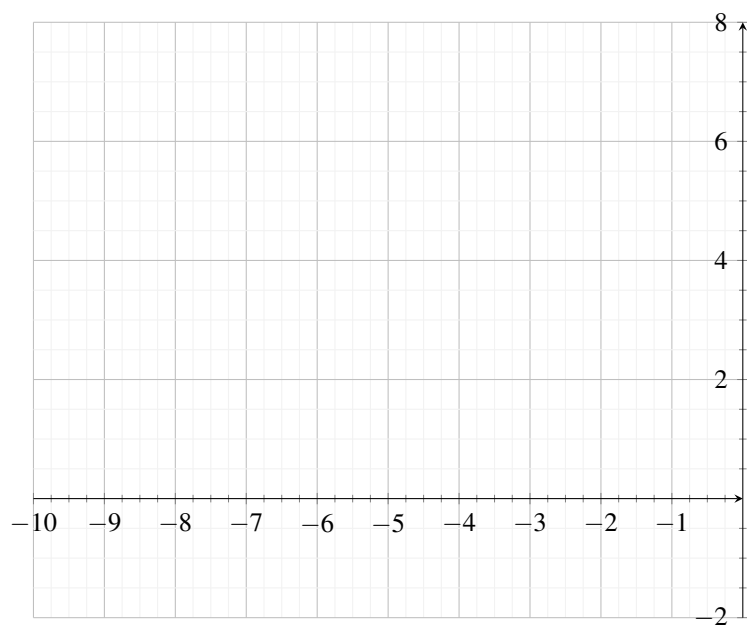
Conic sections

Graph the following conic sections,

1. $x = -8(y-5)^2 - 1$



2. $\frac{(x+5)^2}{4} - \frac{(y-4)^2}{25} = 1$

**Conic Sections**

Standard form of a circle

$$(x-h)^2 + (y-k)^2 = r^2$$

Standard form of a parabola

$$(x-h) = 4a(y-k)^2, \text{ or } (y-k) = 4a(x-h)^2$$

Standard form of an ellipse

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

Standard form of a hyperbola

$$\frac{(x-h)^2}{a^2} - \frac{(y-k)^2}{b^2} = 1$$